

## CYBER SECURITY COURSE

# Windows Persistence and PowerShell Investigation

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## Professional Case Report 03

|                 |                                     |
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| Report date     | 24. septembar 2026                  |
| Case ID         | CSC-2026-03                         |
| Discipline      | Incident response                   |
| Overall risk    | Critical                            |
| Evidence status | Confirmed by saved Redline evidence |

### Purpose

Determine how the attacker maintained execution after reboot and identify PowerShell activity related to archive extraction and follow-on commands.

### Scope note

This report documents work performed in an authorized training environment. Confirmed findings are separated from methodology and expected laboratory behavior.

## Executive Summary

Previously identified attacker files were tested for persistence and correlated with PowerShell event records from 13 June 2017. The Run key connects the malicious file system evidence to a reliable persistence mechanism. PowerShell logs extend the timeline and show how built-in tools supported the intrusion.

## Case Profile

|             |                                                                                                                                  |
|-------------|----------------------------------------------------------------------------------------------------------------------------------|
| Scenario    | Previously identified attacker files were tested for persistence and correlated with PowerShell event records from 13 June 2017. |
| Environment | Windows Redline collection containing Registry, Persistence and Event Log evidence.                                              |
| Tools       | Mandiant Redline, Windows Registry, PowerShell event logs, MITRE ATT&CK mapping                                                  |

## Methodology

I used the following sequence to keep the analysis repeatable and to separate observation from interpretation.

1. Searched Registry and Persistence evidence for known malicious paths.
2. Sorted persistence items by Registry Modified time.
3. Validated file signature status for associated autorun binaries.
4. Filtered PowerShell events between 02:53:47 and 03:14:32 UTC.

## Technical Findings

| ID  | Finding                       | Risk     | Evidence and analysis                                                                                           |
|-----|-------------------------------|----------|-----------------------------------------------------------------------------------------------------------------|
| F01 | Run key persistence           | Critical | C:\ProgramData\kav\svchost.exe was configured under HKLM Software\Microsoft\Windows\CurrentVersion\Run\svchost. |
| F02 | Unsigned autorun target       | High     | The persistence target was not a verified Windows binary despite using the svchost.exe name.                    |
| F03 | PowerShell archive extraction | High     | Event records showed command-line extraction of Tools.zip followed by additional attacker activity.             |

## Representative Commands and Actions

The following commands or analysis actions represent the core of the practical workflow.

Registry filter: Text Value contains C:\ProgramData\kav

PowerShell event time range: 02:53:47 to 03:14:32 UTC

## Risk Assessment

| Risk             | Level    | Operational impact                                                      |
|------------------|----------|-------------------------------------------------------------------------|
| Boot persistence | Critical | Malicious code runs after user logon or restart                         |
| PowerShell abuse | High     | Native tooling reduces malware footprint and may bypass simple controls |

## Recommendations

1. Remove the malicious Run value after forensic preservation.
2. Enable PowerShell Script Block Logging and command-line auditing.
3. Hunt for the same value name, path and hash across endpoints.
4. Use application control to restrict unsigned binaries in ProgramData.

## Knowledge Demonstrated

- Registry forensics
- Persistence analysis
- PowerShell log review
- ATT&CK T1547.001

## Conclusion

The Run key connects the malicious file system evidence to a reliable persistence mechanism. PowerShell logs extend the timeline and show how built-in tools supported the intrusion.

## Hands On Lab Practical 03

Completed as part of the authorized cyber security course practical work.